

Show your work.

1. [3 pts each] Let $\mathbf{x} = (1, 2, 2)$ and $\mathbf{y} = (3, 0, 2)$.

a. Find $\mathbf{x} - \mathbf{y}$.

$$= (1, 2, 2) - (3, 0, 2) = (1 - 3, 2 - 0, 2 - 2) = (-2, 2, 0).$$

b. Find $\mathbf{x} \cdot \mathbf{y}$.

$$= (1, 2, 2) \cdot (3, 0, 2) = (1 \cdot 3, 2 \cdot 0, 2 \cdot 2) = (3, 0, 4)$$

c. Find $\|\mathbf{y}\|$.

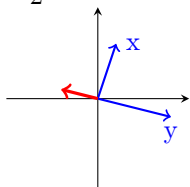
$$= \sqrt{3^2 + 0^2 + 2^2} = \sqrt{13}$$

d. Find a unit vector in the direction of $\mathbf{y}$.

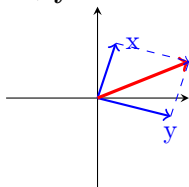
A unit vector has magnitude 1. So we need to scale $\mathbf{y}$ by its own magnitude. $\mathbf{y}/\|\mathbf{y}\| = (3, 0, 2)/\sqrt{13} = (3/\sqrt{13}, 0, 2/\sqrt{13})$.

2. [3 pts each] Use the provided graph to sketch the following vectors.

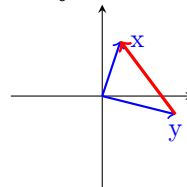
a. $-\frac{1}{2}\mathbf{y}$



b. $\mathbf{x} + \mathbf{y}$



c. $\mathbf{x} - \mathbf{y}$



3. [5 pts each] Give Cartesian equations of the given hyperplanes.

a. The hyperplane in $\mathbb{R}^2$ passing through $(1, 1)$ and orthogonal to the line $\mathbf{x} = (0, 2) + t(2, 1)$

It is orthogonal to the given line so $\mathbf{a} = (2, 1)$ is a normal vector.

$$\begin{aligned} \mathbf{a} \cdot \mathbf{x} &= \mathbf{a} \cdot \mathbf{x}_0 \\ (2, 1) \cdot (x_1, x_2) &= (2, 1) \cdot (1, 1) \\ 2x_1 + x_2 &= 3 \end{aligned}$$

b. The hyperplane in $\mathbb{R}^3$ which contains the point $(0, 1, 0)$ and has normal vector $(1, 2, 1)$.

$$\mathbf{a} = (1, 2, 1)$$

$$\begin{aligned} \mathbf{a} \cdot \mathbf{x} &= \mathbf{a} \cdot \mathbf{x}_0 \\ (1, 2, 1) \cdot (x_1, x_2, x_3) &= (1, 2, 1) \cdot (0, 1, 0) \\ x_1 + 2x_2 + x_3 &= 2 \end{aligned}$$

4. Consider the matrix,

$$A = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 2 & 0 & 0 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

a. [4 pts] Find the reduced echelon form of A .

$$A = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 2 & 0 & 0 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix} \rightsquigarrow \begin{matrix} 1/2R2 \\ R1 \\ R3 \end{matrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix} \rightsquigarrow \begin{matrix} 1/2(R3-R2) \end{matrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1/2 \end{bmatrix}$$

b. [4pts] Give the general solution of $A\mathbf{x} = \mathbf{0}$ in *standard* form.

$x_1 + x_4 = 0$, $x_2 + 2x_4 = 0$, and $x_3 - 1/2x_4 = 0$. So $\mathbf{x} = (-x_4, -2x_4, 1/2x_4, x_4) = x_4(1, 2 - 1/2, 1)$ or

$$\mathbf{x} = x_4 \begin{bmatrix} -1 \\ -2 \\ 1/2 \\ 1 \end{bmatrix}$$

- c. [2 pts] Is the reduced echelon form of any matrix unique? i.e., if two different people compute the reduced echelon form of the same matrix, will they always arrive at the exact same result?

YES.

5. Consider the system of equations

$$\begin{array}{rcl} x_1 & +x_3 & = 3 \\ x_2 & +2x_3 & = 1 \\ 2x_2 & +4x_3 & = 2 \end{array}$$

- a. [2 pts] Express the system as an augmented matrix.

$$[A\mathbf{x}|\mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 2 \end{array} \right]$$

- b. [4 pts] Find the reduced echelon form of this augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 2 \end{array} \right] \rightsquigarrow \text{R3} - 2\text{R2} \quad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

- c. [4 pts] Give the general solution to the system of equations.

$x_1 + x_3 = 3$ and $x_2 + 2x_3 = 1$. So then $\mathbf{x} = (3 - x_3, 1 - 2x_3, x_3) = (3, 1, 0) + x_3(-1, -2, 1)$ or

$$\mathbf{x} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$$

6. [3 pts each] Consider the matrices

$$A = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

- a. How many solutions does the system of equations represented by $A\mathbf{x} = \mathbf{b}$ have?

None.

- b. What is the *rank* of A ?

It has one row of zeros. So the $\text{rank}(A) = 2$ (number of rows minus 1).

- c. How many constraints does the solutions of $A\mathbf{x} = \mathbf{b}_1$ have if

$$\mathbf{b}_1 = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

It will have 1 constraint equation since it has 1 row of zeros.

7. Complete the following

- a. [2 pts] Suppose that the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ span the hyperplane $\mathcal{H}$, and $\mathbf{w} = 2\mathbf{v}_1 - \mathbf{v}_2 + 3\mathbf{v}_3$. Is $\mathbf{w}$ in $\mathcal{H}$?

YES. Since $\mathbf{w}$ is a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

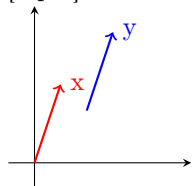
- b. [2 pts] Suppose that the $n \times n$ matrix A is nonsingular. Does $A\mathbf{x} = \mathbf{b}$ have a solution? If so, is it unique? (Recall that nonsingular means that the reduced echelon form of the matrix does not have a row with all zeros.)

YES it has a solution and YES it is unique. If A is nonsingular then its reduced echelon form will have a diagonal of 1's with zeros elsewhere. So the augmented matrix will look like,

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 1 & b_n \end{array} \right]$$

So the solution will be $x_1 = b_1, x_2 = b_2, \dots, x_n = b_n$.

- c. [2 pts] Let $\mathbf{x}$ and $\mathbf{y}$ be vectors shown in the figure below. Does $\mathbf{x} = \mathbf{y}$?



YES.

- d. [4 pts] Let

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}.$$

Find $A + B$.

$$A + B = \begin{bmatrix} 0+2 & 1+2 \\ 2+2 & 3+2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

8. [Bonus] Prove that $n \times n$ matrices have an identity element under matrix addition and multiplication, i.e., that given any $n \times n$ matrix A there exists a matrix B and C such that $A + B = A$ and $AC = A$.

Proof. Let

$$B = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \text{ and } C = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

(This is the zero matrix, $\mathbf{0}$, and the identity matrix I for $n \times n$ matrices respectively.) So then,

$$A + B = A + \mathbf{0} = \begin{bmatrix} a_{11} + 0 & a_{12} + 0 & \cdots & a_{1n} + 0 \\ a_{21} + 0 & a_{22} + 0 & \cdots & a_{2n} + 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} + 0 & a_{n2} + 0 & \cdots & a_{nn} + 0 \end{bmatrix} = A$$

For $AC = AI$ we have that the entry for the ij^{th} place is $(AC)_{ij} = (a_{i1}, a_{i2}, \dots, a_{in}) \cdot (c_{j1}, c_{j2}, \dots, c_{jn})$. But $c_{jk} = 0$ for every coordinate except $c_{jj} = 1$. So then

$$(AC)_{ij} = (a_{i1}, a_{i2}, \dots, a_{in}) \cdot (0, 0, \dots, 1, \dots, 0) = a_{i1} \cdot 0 + a_{i2} \cdot 0 + \cdots + 1 \cdot a_{ij} + \cdots + 0 \cdot a_{in} = a_{ij}$$

Thus $AC = AI = A$ (note also that it similarly follows that $\mathbf{0} + A = A$ and $IA = A$; it is usually required that identity elements be commutative.) $\square$